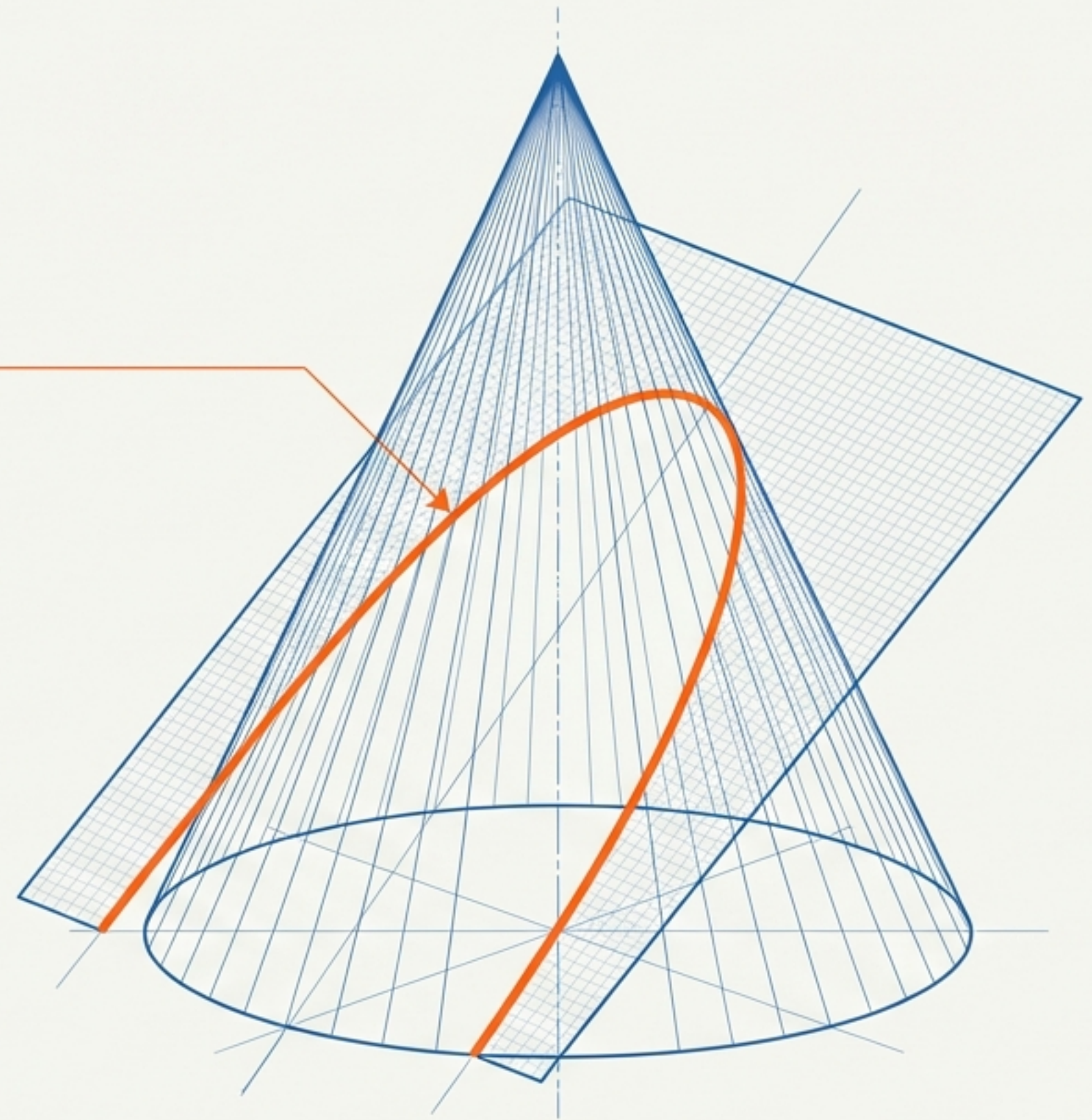


PARABOLA

The Complete JEE Mastery Guide

From Conic Definitions to Advanced Shortcuts & Properties

Comprehensive Revision Notes • Based on Arjuna JEE Series



The Foundation: Conic Sections

Definition:

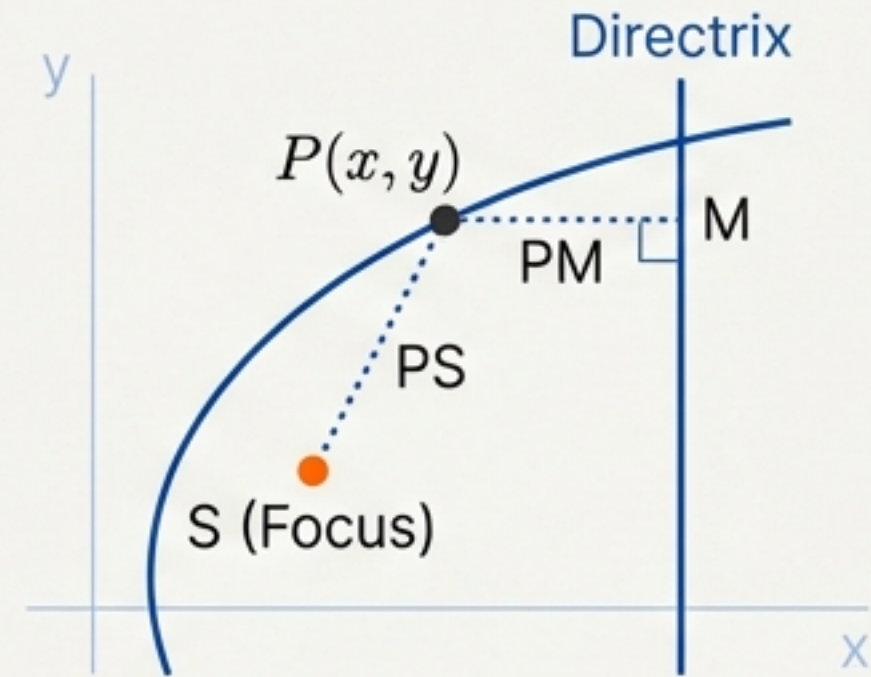
The locus of a point P such that the ratio of its distance from a Fixed Point (Focus) to a Fixed Line (Directrix) is constant.

The Golden Rule:

$$\frac{PS}{PM} = e \text{ (Eccentricity)}$$

S = Focus

M = Perpendicular projection on Directrix



Condition	Curve Name
$e = 1$	Parabola
$0 < e < 1$	Ellipse
$e > 1$	Hyperbola

Aside

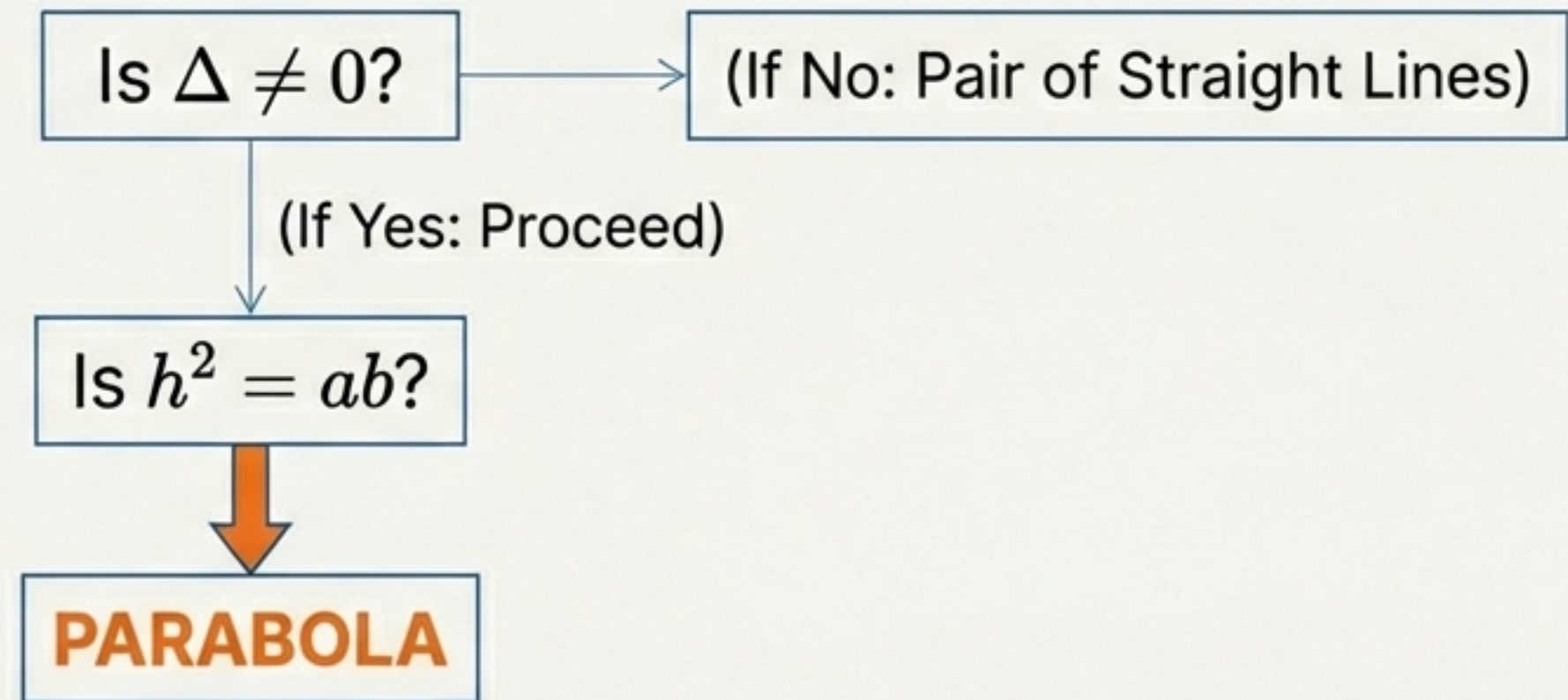
Memory Aid:

Why was the Ellipse silent? Because the others were **Parabola** and **Hyperbola**.
(Hint: "Bola" implies speaking).

Recognizing a Parabola

Analyzing the General Equation of Second Degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$



Pro-Tip: The Eye Test

Example A: $(2x + 3y)^2 + 5x = 0$ ✓

2nd degree terms form a Perfect Square → **Parabola**

Example B: $x^2 + y^2 + x = 0$ ✗

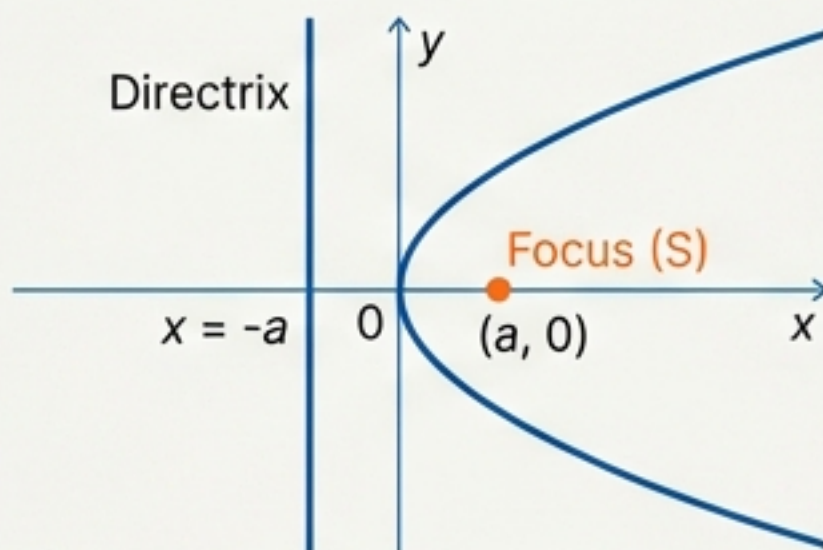
No xy term, ont, coeff $x^2 = y^2$ → **Circle**

Standard Equations: The Bible

Memorize the Standard (y^2), derive the Upward (x^2).

Right-Handed Parabola

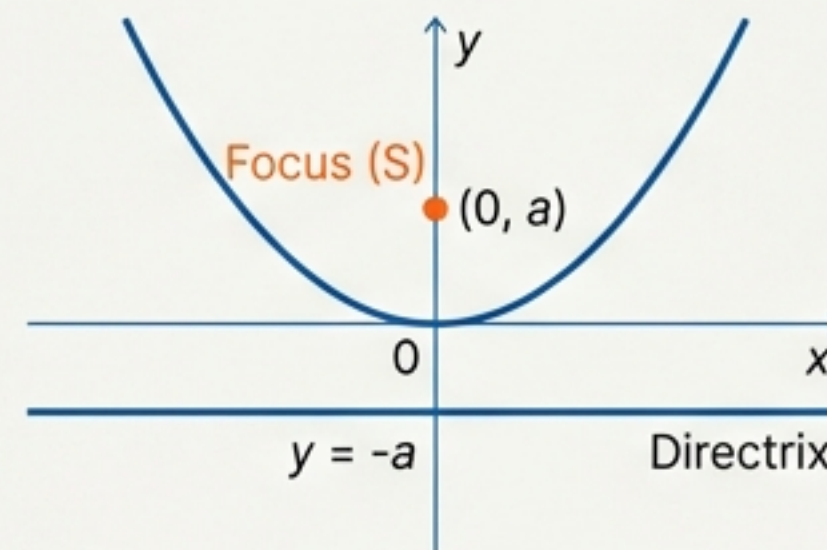
$$y^2 = 4ax$$



Parameter	Value
Focus	$(a, 0)$
Directrix	$x = -a$
Vertex	$(0, 0)$
Axis	$y = 0$ (X-axis)
Latus Rectum	$4a$

Upward Parabola

$$x^2 = 4ay$$

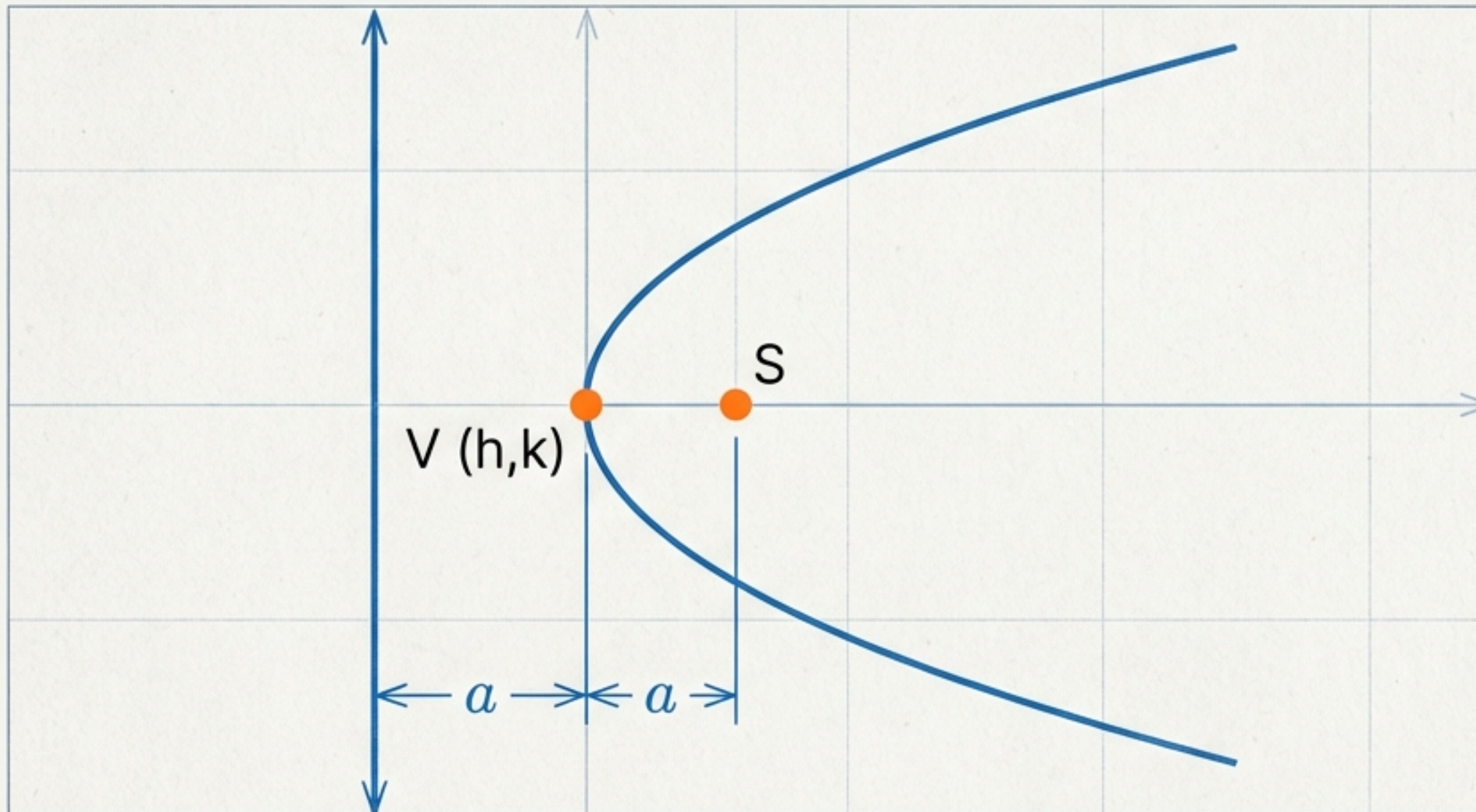


Parameter	Value
Focus	$(0, a)$
Directrix	$y = -a$
Vertex	$(0, 0)$
Axis	$x = 0$ (Y-axis)
Latus Rectum	$4a$

Shifted Parabolas

When the Vertex moves to (h, k)

$$(y - k)^2 = 4a(x - h) \quad \text{New Vertex } V = (h, k)$$



****The Distance Mindset****

1. Standardize using Completing the Square.
2. Ignore coordinates initially. Think in Distances.
3. Vertex to Focus distance = a .
4. Vertex to Directrix distance = a .
5. Focus to Directrix distance = $2a$.

Parametric Coordinates: The Power of '\$t\$'

The "Trigonometry" of Coordinate Geometry. Reduces two variables (x, y) to one (t) .

The Substitution

- For $y^2 = 4ax \rightarrow$ Point $P(t) = (at^2, 2at)$
- For $x^2 = 4ay \rightarrow$ Point $P(t) = (2at, at^2)$



Focal Distance Formula

Definition: Distance of point P from Focus S .

$$\text{Formula: } PS = a + at^2$$

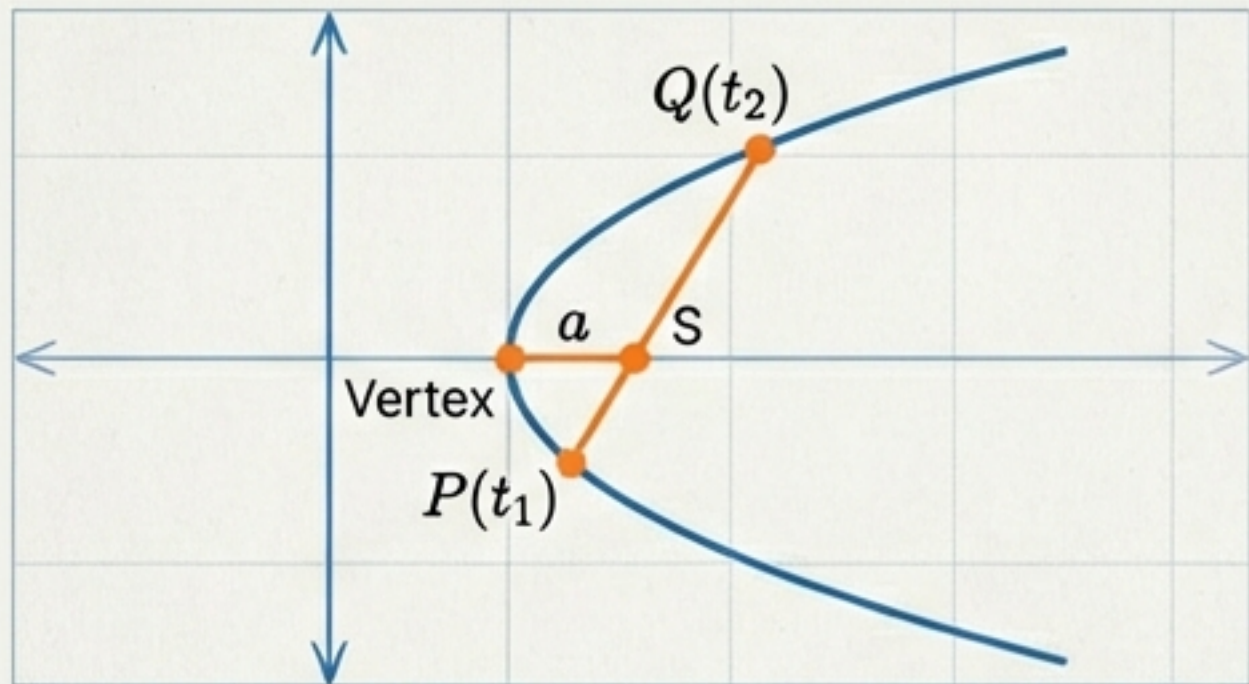
(Small note: "For $y^2 = 4ax$ ")

When to use:

Always start difficult problems involving chords, tangents, or normals by assuming the point $P(t)$.

Focal Chords

A chord passing through the Focus $(a, 0)$.



The Golden Condition

If a chord joins t_1 and t_2 and passes through the focus:

$$t_1 t_2 = -1$$

Length & Properties

Parametric Length

$$L = a \left(t - \frac{1}{t} \right)^2$$

Trigonometric Length

$$L = 4a \csc^2 \theta$$

$(\theta = \text{angle with axis})$

Harmonic Mean

The Semi-Latus Rectum $(2a)$ is the Harmonic Mean of segments SP and SQ .

$$\text{Formula: } \frac{2(SP)(SQ)}{SP + SQ} = 2a$$

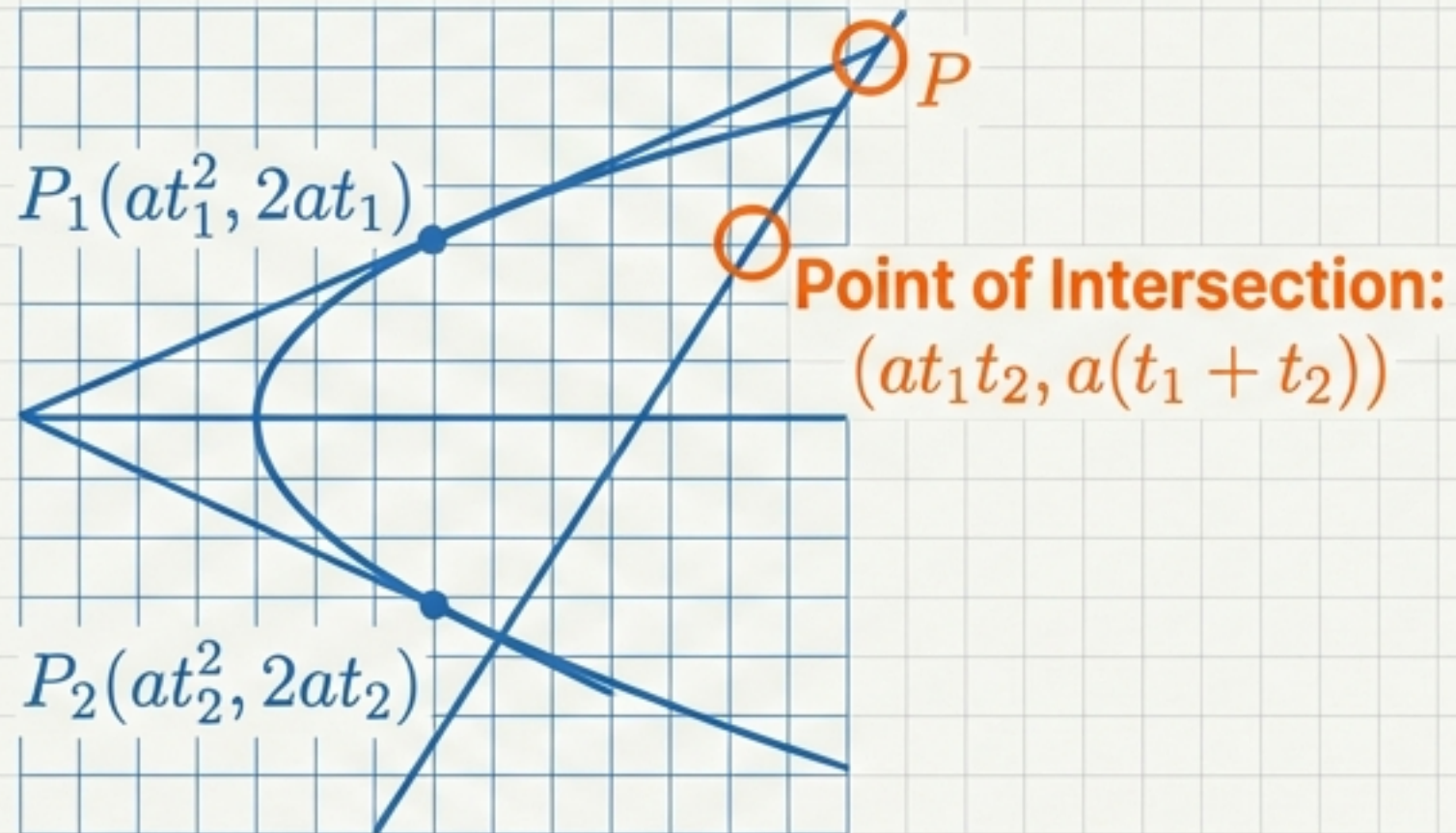
Tangents: Equations & Forms

Reference Table for $y^2 = 4ax$

Form	Equation	Context & Notes
Point Form ($T = 0$)	$yy_1 = 2a(x + x_1)$	Use when point (x_1, y_1) is given on the curve.
Parametric Form (The MVP)	$ty = x + at^2$	Slope $m = 1/t$. Memorize this. Solves 70% of tangent questions.
Slope Form	$y = mx + \frac{a}{m}$	Condition of Tangency: $c = \frac{a}{m}$. Note: Specific slope m produces only ONE tangent (No parallel tangents) .

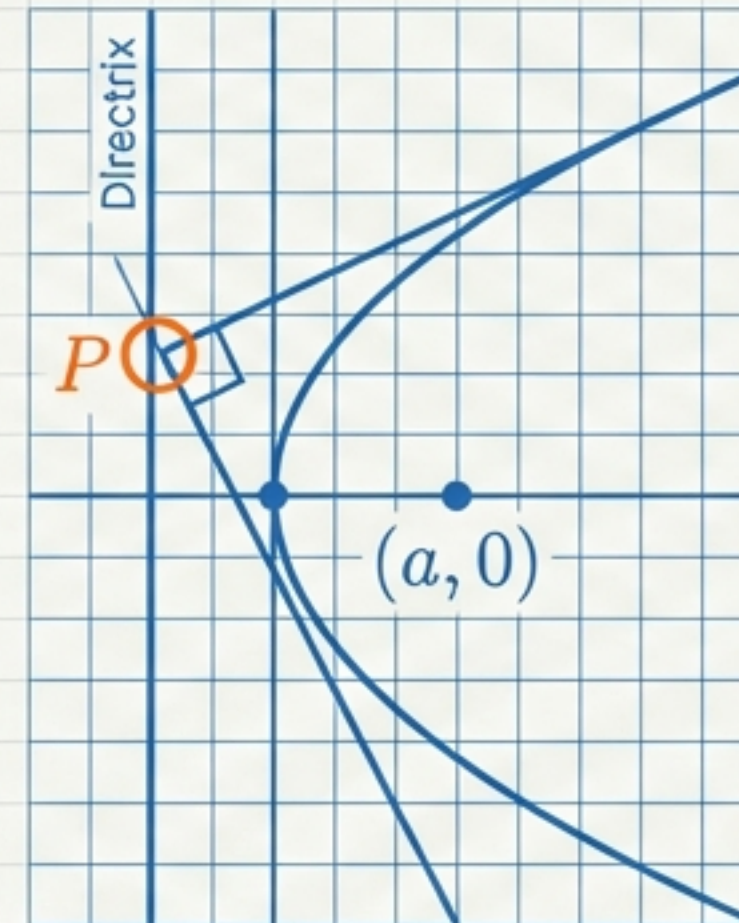
Tangents: Intersections & Properties

General Intersection



Mnemonic: Geometric Mean of x-coords,
Arithmetic Mean of y-coords.

Special Properties



The Director Property:
Perpendicular tangents
always intersect on the
Directrix.

Focal Chord Property:
Tangents at the ends of
a focal chord intersect
at 90° on the Directrix.

Caution! There is no Director Circle for a Parabola.
It is a Director Line (The Directrix).

Algorithm: Common Tangents

Solving for lines touching two curves (e.g., Parabola + Circle)

1

Step One

Write the **Slope Form** equation for the first curve.

For Parabola: $y = mx + \frac{a}{m}$

2

Step Two

Apply the condition of tangency for the second curve.

- If Circle: Distance from Center = Radius ($p = r$).
- If Parabola: Compare coefficients.

3

Step Three

Solve for m .

Golden Rule: Always use the Slope Form ($y=mx+c$). Never use the $T=0$ point form for common tangents.

Normals: The Perpendiculars

Equations & Re-intersection

The Equations

Slope Form (The “Beast”):

$$y = mx - 2am - am^3$$

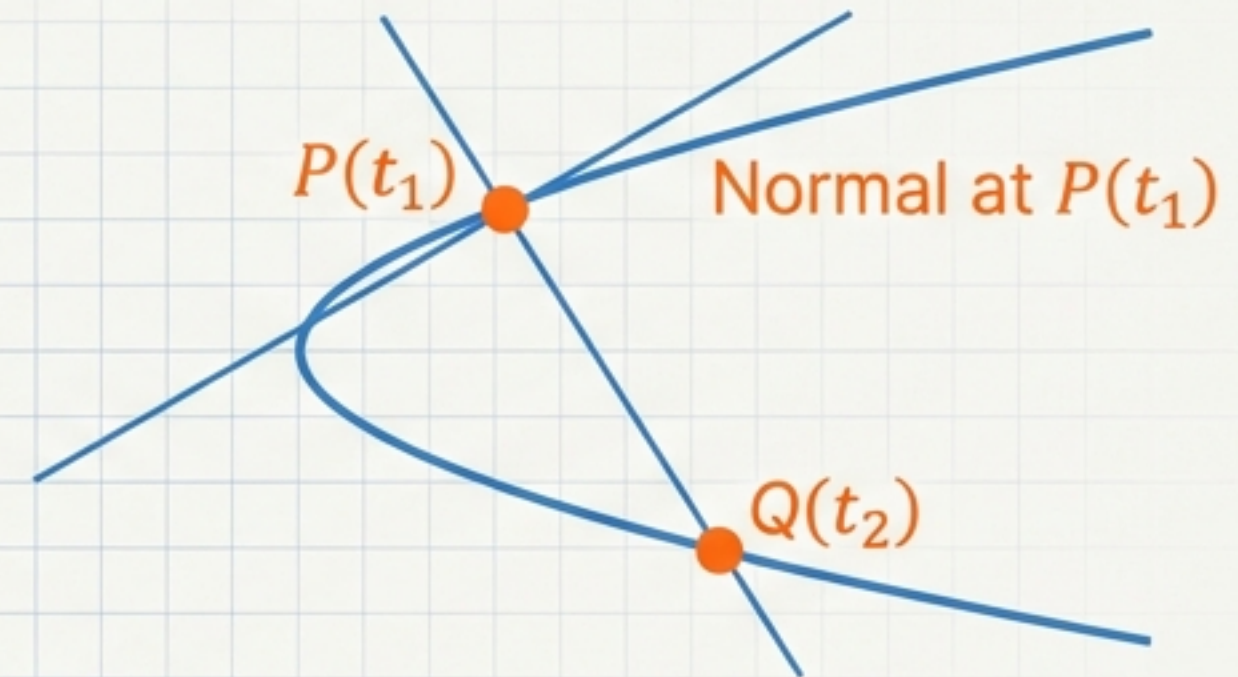
Note: Cubic in $m \rightarrow$ Up to 3 normals from one point (h, k) .

Parametric Form:

$$y + tx = 2at + at^3$$

Slope relation: $m_{normal} = -t$

Re-intersection



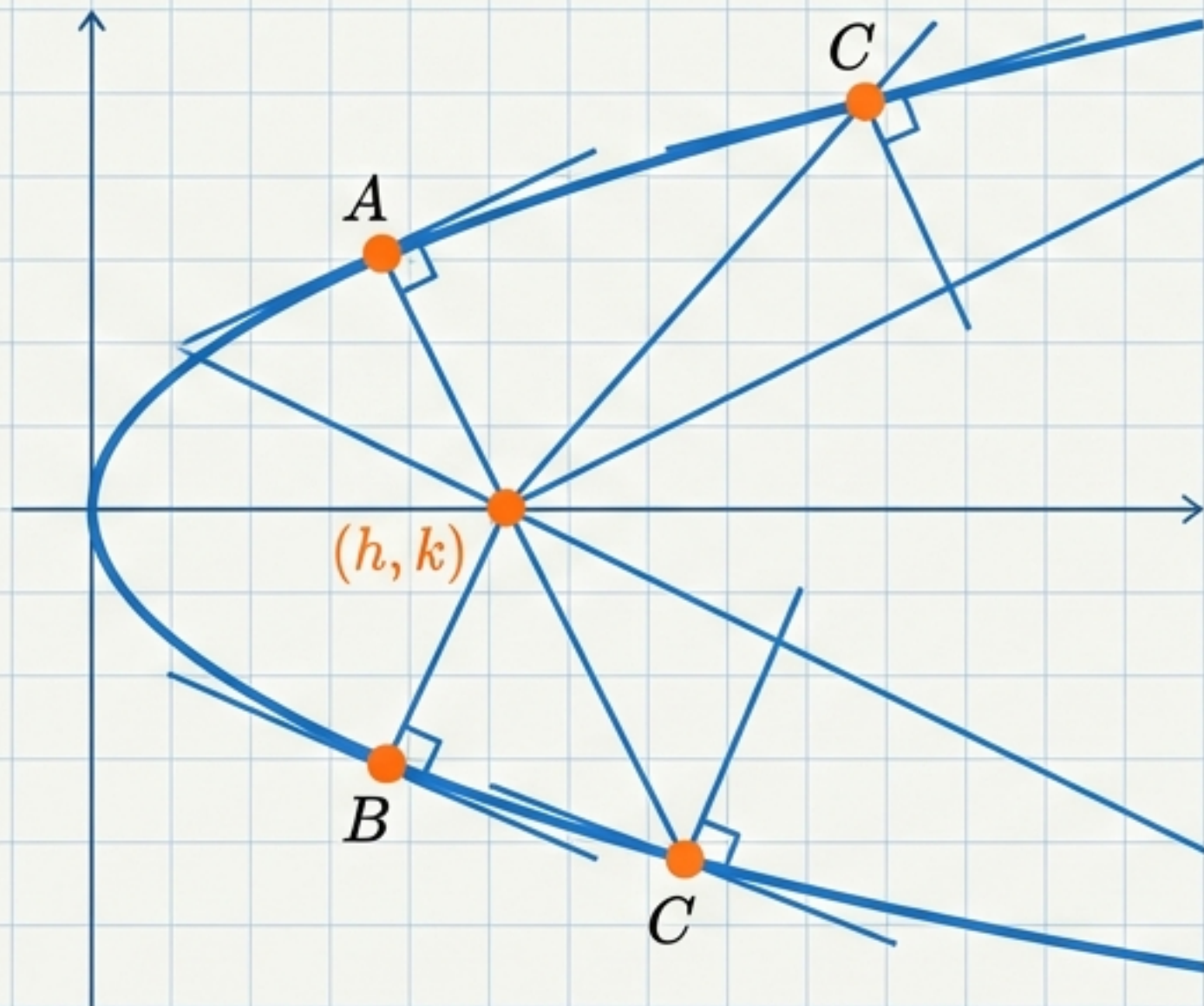
If normal at t_1 meets curve again at t_2 :

$$t_2 = -t_1 - \frac{2}{t_1}$$

Pro-Tip: $t_2^2 \geq 8$

Co-Normal Points

The feet of the three normals drawn from a single point (h, k) .



Properties

1. Sum of Ordinates

If parameters are t_1, t_2, t_3 ,
then $\sum t = 0$.

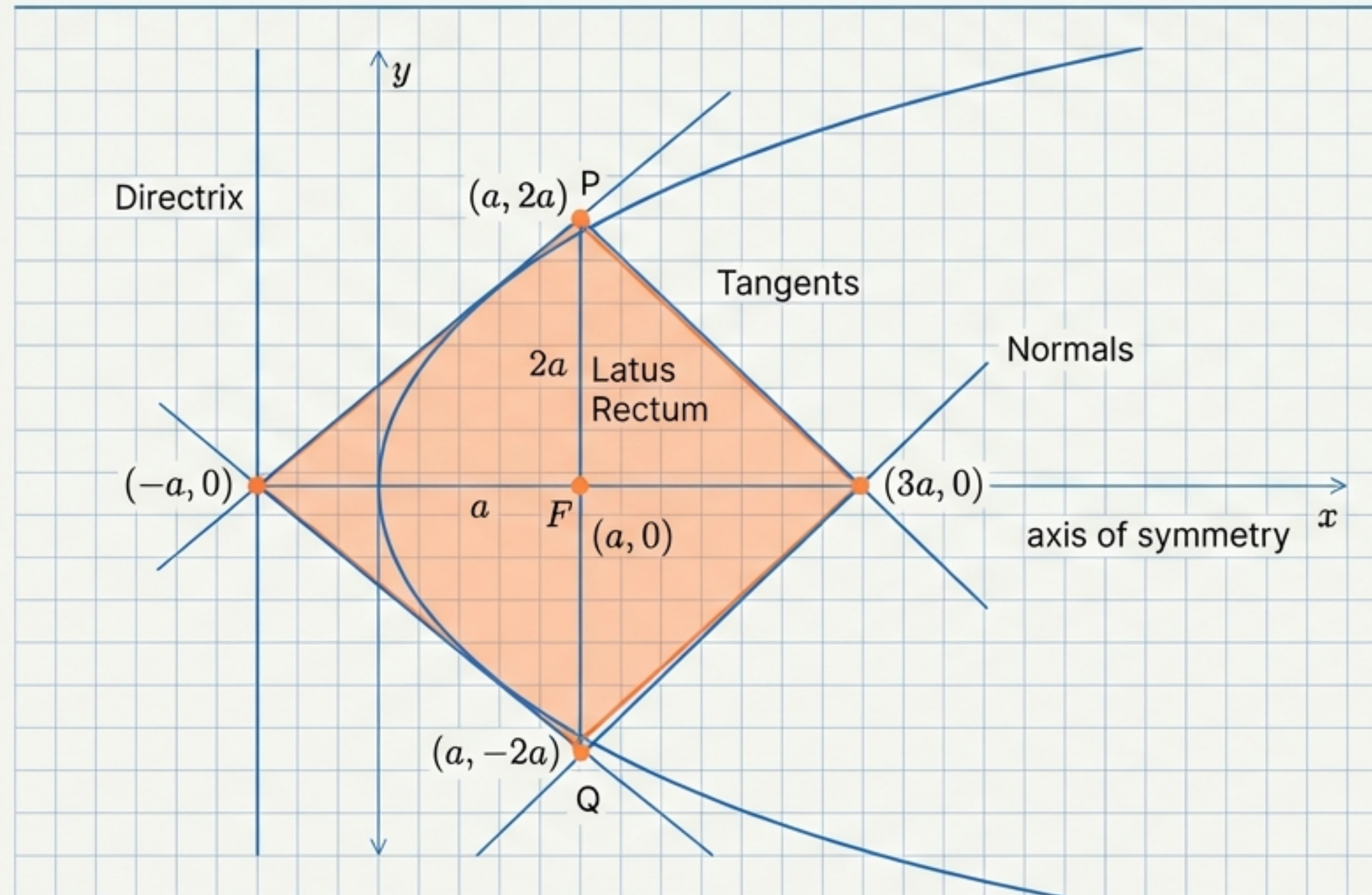
$$\implies y_1 + y_2 + y_3 = 0$$

2. Centroid Property

The Centroid of the triangle formed by co-normal points (ABC) lies on the **Axis** of the parabola.

The Square Property

Shortcut for Latus Rectum Geometry



Data Points

- **Shape:** Square
- **Side Length:** $2\sqrt{2}a$
- **Tangents Meet:** On Directrix $(-a, 0)$
- **Normals Meet:** On Axis $(3a, 0)$

Pro-Tip: The Latus rectum forms a rotated square which has a Safety Orange underside, and the two corners are Latin Modern Math.

Exam Strategy: The Decision Matrix

Which tool solves which problem?

Scenario / Given Info	Recommended Tool
Point given ON the curve (x_1, y_1)	$T = 0$ (Tangent) or $ty = x + at^2$
Slope (m) is given	$y = mx + a/m$
Intersection of lines / Chords	Parametric coords (t)
Focal Chord mentioned	$t_1 t_2 = -1$
Common Tangent	Compare Slope Forms
Shifted Parabola	Calculate Distances (a) from Vertex

Final Mastery Checklist

Do you memorize...? in Inter

- ☐ Standard Table ($y^2 = 4ax$ focus, directrix, etc.)
- ☐ Tangent Equation ($ty = x + at^2$)
- ☐ Normal Equation ($y + tx = 2at + at^3$)
- ☐ Focal Chord Length ($a(t - 1/t)^2$)
- ☐ Co-normal property ($\sum y = 0$)



Coordinate Geometry is a game of memory and speed. Don't derive in the exam—apply the result.

Practice Goal: Solve 20 Common Tangent & Normal problems using these shortcuts today.